

Solvesheet

BRAC University (Department of Computer Science and Engineering)  
CSE 320 (Data Communication) | Summer 2023

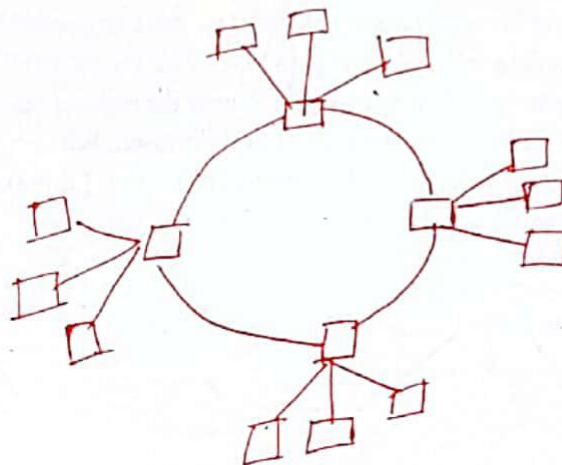
Quiz 1 (Set A)

Full Marks: 15

Duration: 20 minutes

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

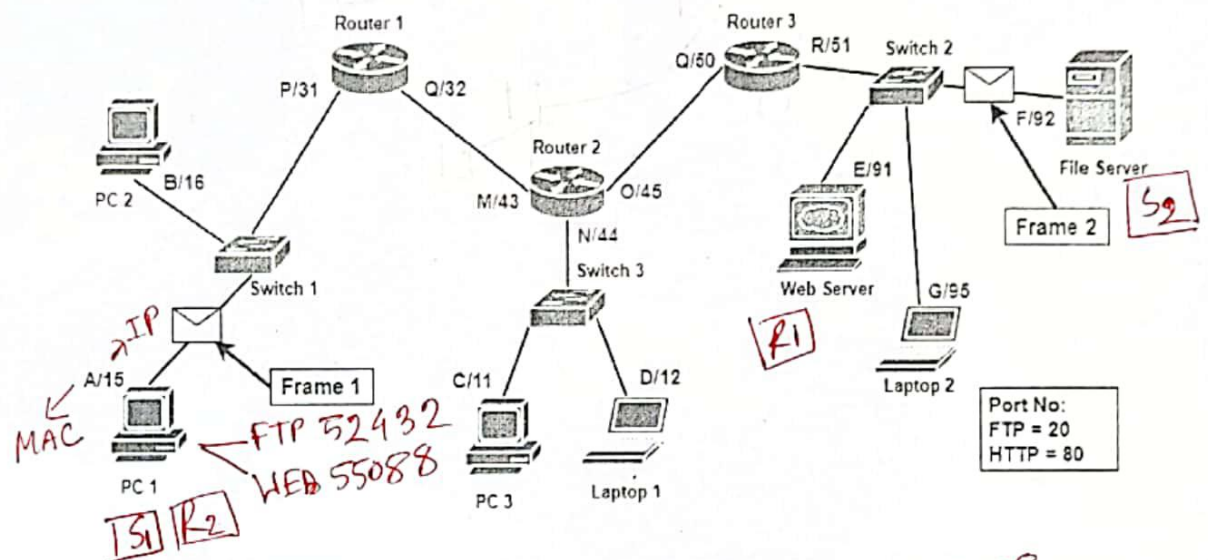
1. Illustrate diagrammatically a hybrid topology with a ring backbone and four star networks consisting of 3 nodes at each hub. In the topology drawn, identify one possible problem or failure that could bring the whole network down and justify your answer. [ 3 marks ]



2. Match the following to one or more layers of the OSI model: [ 6 marks ]

- a. Node to node error control → Data Link
- b. Source to destination delivery of packets → Network
- c. Compression of message → Presentation
- d. Identification of different application processes → Transport
- e. Interface to transmission media → Physical
- f. Access control over a link → Data link

3. Label the frames (1 & 2) shown below with appropriate port, IP and MAC addresses. The sender Host PC1 has two applications running; one for FTP with port number 52432 and the other for accessing the web server with port number 55088. Frame 1 is intended for the web server from PC1 and Frame 2 is coming from file server to PC1. Web server and File server use port number 80 and 20 respectively to send data to clients. Assume the numbers mentioned in the figure represent IP addresses and the alphabets are MAC addresses. While mentioning the port addresses, you can use any value from the appropriate ranges. [ 6 marks ]



|      |         |          |          |         |         |              |              |
|------|---------|----------|----------|---------|---------|--------------|--------------|
| web  | Frame 1 | S. IP 15 | D. IP 91 | S.MAC A | D.MAC P | S.Port 55088 | D.Port 80    |
| File | Frame 2 | S. IP 92 | D. IP 15 | S.MAC F | D.MAC R | S.Port 20    | D.Port 52432 |

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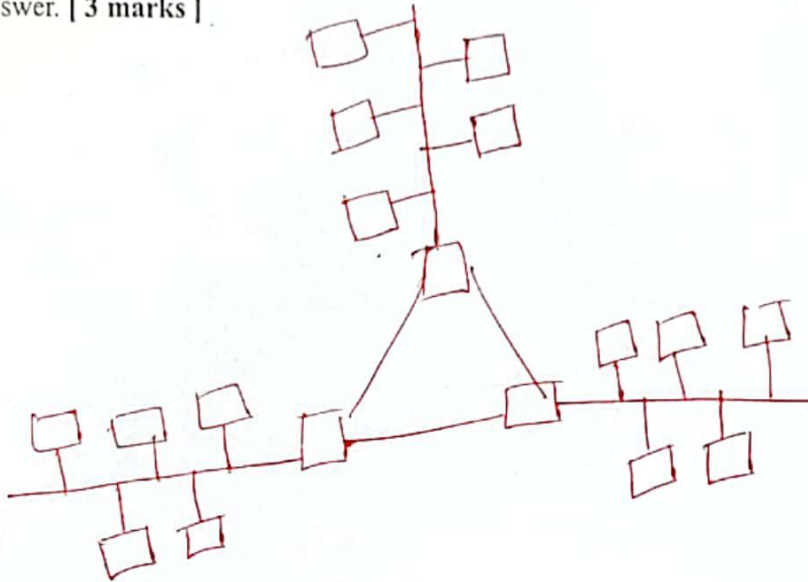
Quiz 1 (Set B)

Full Marks: 15

Duration: 20 minutes

|       |     |          |
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| Name: | ID: | Section: |
|-------|-----|----------|

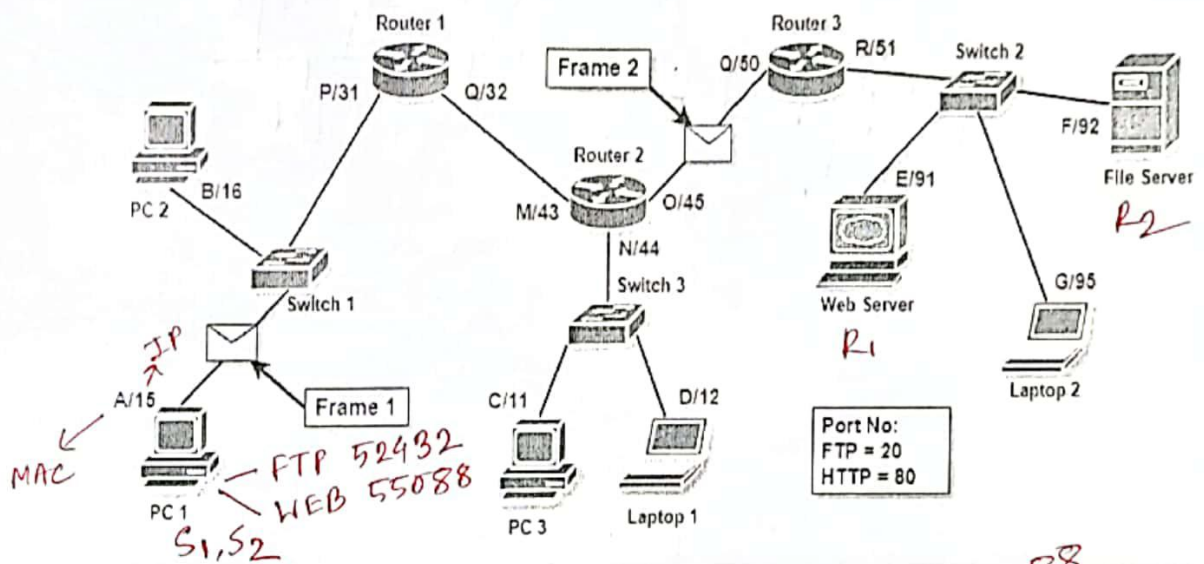
1. Illustrate diagrammatically a hybrid topology with a mesh backbone and three bus networks connected to it. Each bus network consists of five nodes. In the topology drawn, identify one possible problem or failure that could lead to degraded network performance and justify your answer. [ 3 marks ]



2. Match the following to one or more layers of the TCP/IP model: [ 6 marks ]

- a. Restarting browsers which was idle for a long time → Application
- b. Error control and retransmission of segments of data → Transport
- c. Transmission of bit stream across physical medium → physical / Network Access
- d. Direct communication with user's application program → Application
- e. Access control over a link → Data link / Network Access
- f. Source to destination delivery of packets → Internet

3. Label the frames (1 & 2) shown below with appropriate port, IP and MAC addresses. The sender Host PC1 has two applications running; one for FTP with port number 52432 and the other for accessing the web server with port number 55088. Frame 1 is intended for the web server and Frame 2 is intended for the file server from PC1. Web server and File server use port number 80 and 20 respectively to send data to clients. Assume the numbers mentioned in the figure represent IP addresses and the alphabets are MAC addresses. While mentioning the port addresses, you can use any value from the appropriate ranges. [ 6 marks ]



|      |         |          |          |         |         |              |           |
|------|---------|----------|----------|---------|---------|--------------|-----------|
| web  | Frame 1 | S. IP 15 | D. IP 91 | S.MAC A | D.MAC P | S.Port 55088 | D.Port 80 |
| file | Frame 2 | S. IP 15 | D. IP 92 | S.MAC 0 | D.MAC Q | S.Port 52432 | D.Port 20 |